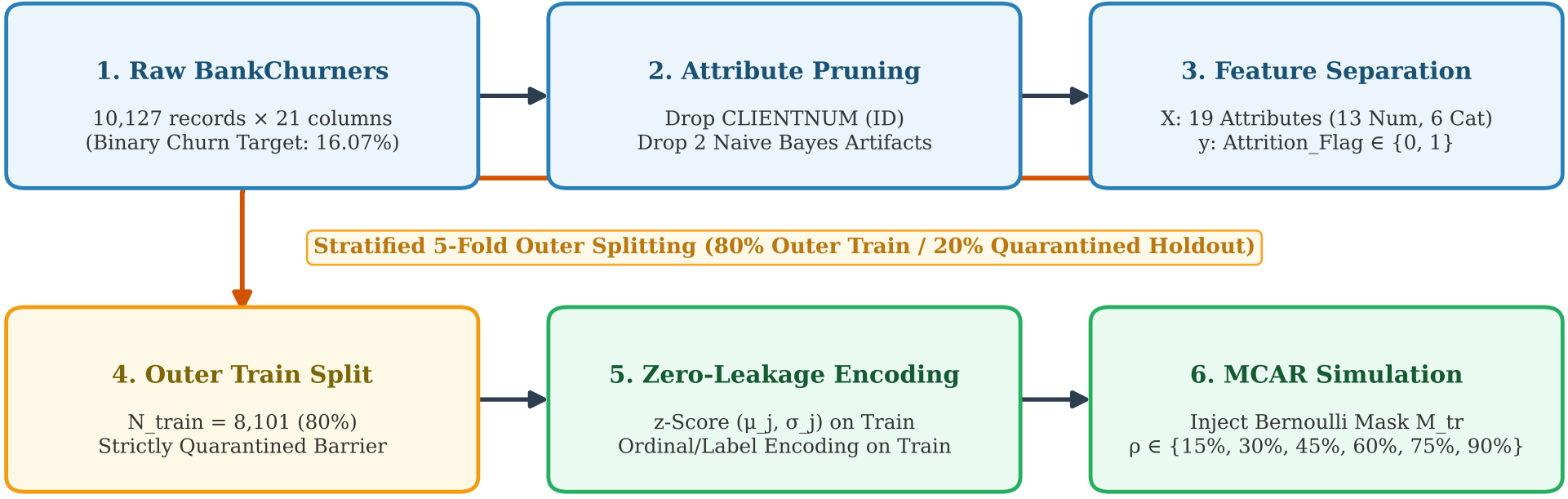
1. Raw BankChurners
10,127 records × 21 columns
(Binary Churn Target: 16.07%)
2. Attribute Pruning
Drop CLIENTNUM (ID)
Drop 2 Naive Bayes Artifacts
3. Feature Separation
X: 19 Attributes (13 Num, 6 Cat)
y: Attrition_Flag ∈ {0, 1}
Stratified 5-Fold Outer Splitting (80% Outer Train / 20% Quarantined Holdout)
4. Outer Train Split
N_train = 8,101 (80%)
Strictly Quarantined Barrier
5. Zero-Leakage Encoding
z-Score (μ_j, σ_j) on Train
Ordinal/Label Encoding on Train
6. MCAR Simulation
Inject Bernoulli Mask M_tr
ρ ∈ {15%, 30%, 45%, 60%, 75%, 90%}